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Received: 30 December 2020 | Revised: 29 April 2021 | Accepted: 3 June 2021

DOI: 10.1002/acs.3297

SPECIAL ISSUE ARTICLE

WILEY

Robust attitude estimation of rotating space debris based on virtual observations of neural network

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Summary

High precise estimation and prediction of the target's attitude motion are key technologies for capturing and removing rotating space debris. In this article, a neural-network-enhanced Kalman filter (NNEKF) is proposed to improve the precision and robustness of attitude estimation algorithm. The main innovation of the NNEKF is to utilize virtual observations of the inertia characteristics to improve the filter's performances. The virtual observations are obtained using a neural network, which is offline trained using simulation data. In order to decrease the number of nodes of the network, the input data are preprocessed using the discrete Fourier transformation method. Moreover, by involving the characteristic frequencies in the input vector, the neural network can extract information from all the past observations, so as to grasp long-term characteristics of the dynamical system. Therefore, the NNEKF can provide more precise estimation of the target's moment of inertia, and furthermore improve the accuracy and robustness of attitude estimation and prediction. Simulation results indicate that the NNEKF can reduce the estimation errors by 39% compared with the conventional EKF method when using the same measurement data. And the accumulation errors of prediction using estimates of the NNEKF is just as 24% as the conventional EKF.

KEYWORDS

adaptive attitude estimation, information fusion, neural network, space debris

1 | INTRODUCTION

Along with the development of space explorations and operations, the number of space debris has increased rapidly in past decades.[1] In many scenarios, the space debris is considered as one of the biggest threats for spacecrafts in orbit.[2] Consequently, technologies for capturing and removing the space debris have attracted many studies in recent years. For now, the most effective way to remove the heavy space debris, like the inactive satellite and the discarded rocket booster, is to capture it using a manipulator or other equipment for deorbiting,[3,4] as shown in Figure 1. Since the disturbing forces acting on the debris, such as air drags, is very small, most of space debris will keep rotating under the effect of the initial angular velocity.[5] Hence it is necessary to take an accuracy estimation and prediction for the target's attitude motion before capturing, to avoid crashes and damages during operations.

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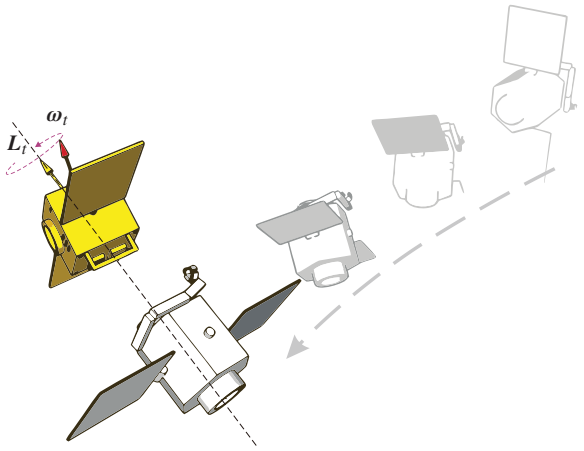


FIGURE 1 A servicing satellite (white) trying to capture a rotating space debris (yellow)

In practice, the servicing satellite cannot know the dynamical parameters of the space debris in advance, such as the mass and the moment of inertia. In addition, the servicing satellite cannot measure the debris' angular velocity directly, but can only obtain its attitude quaternion with respect to the serving satellite using noncontacted instruments, like version systems[6,7] and Light Detection and Ranging (LIDAR) systems.[8-10] Based on the rough measurements, the servicing satellite can estimate the debris' angular velocity and inertia tensor via nonlinear filters, and further predict its attitude motion using numerical integration methods.

Since the moment of inertia can effect the second-order derivative of attitude quaternions, it has a relatively large long-term effect on the target's angular velocity. Consequently, it is necessary to get good estimates of the target's moment of inertia for precise prediction of the attitude motion. The extend Kalman filter (EKF) is a widely used method for attitude estimations of the spacecraft.[11] However, it is hard for the EKF to obtain a very precise estimation of the debris' moment of inertia mainly for three reasons: 1) the measurement errors of LIDAR and version systems are relatively large (commonly more than 10 degrees); 2) both LIDAR and version systems need complex algorithms which cost much time for each measurement update, hence the measurement frequency is respectively low (commonly less than 2 Hz); and 3) the moment of inertia has a very small effect on the attitude motion in a short time, hence the moment of inertia has a low observability for the filter, making it hard to converge to the truth value by just using the attitude measurements. Therefore, the prediction of the attitude quaternion and the angular velocity will rapidly diverge from the truth value when no measurement is available to update. The development of nonlinear filters, such as high-order extended Kalman filter (HEKF), unscented Kalman filter (UKF), cubature Kalman filter (CKF), and particle filter (PF), can solve the measurement error and low measurement frequency problems to a certain extent.[12] However, the improvement of the filter cannot solve the low observability problem, which is one of the essential problems of the recursive estimation algorithm.

Fortunately, the development of artificial intelligence methods, such as the neural network (NN) technology, provides an alternative way to estimate the target's moment of inertia accurately. For example, Hou et al. designed a neural network to predict the attitude motion of the spacecraft when the measurements break off.[13] This method needs online training of the network, leading to large computational burdens in practical applications. Ma et al. designed a neural network and associated preprocessing algorithm to estimate the rotating space debris' inertia parameters.[14] However, it does not involve the target's attitude dynamics to further enhance the estimation accuracy. Al-Sharman et al. designed a deep learning framework for attitude estimation to identify the measurement noise models and filter them out, where the target's inertia parameters are assumed a priori known.[15]

Based on the previous studies, this article provides a neural-network-enhanced Kalman filter (NNEKF) to improve the precision and robustness of attitude estimation and prediction of rotating debris. The main innovation of the NNEKF is to utilize virtual observations of the inertia characteristics to improve the filter's performances, as shown in Figure 2. The virtual observations are obtained using a neural network, which is offline trained using simulation data. The input data are preprocessed using the discrete Fourier transformation (DFT) method, so as to decrease the number of nodes of the network. In Figure 2, the flow path colored in red represents the conventional EKF method for attitude estimation and

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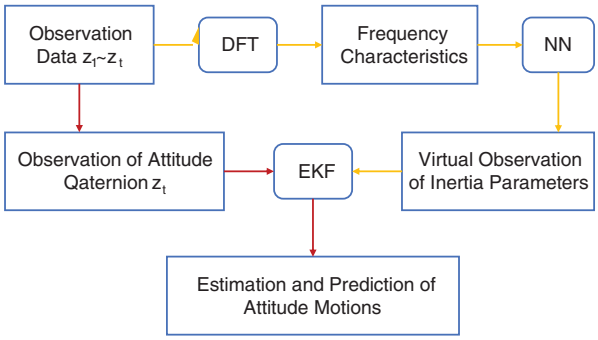


FIGURE 2 Flow diagram of the neural-network-enhanced Kalman filter (NNEKF) based on virtual observations

prediction of the space debris, leading to large estimation errors of the moment of inertia and large prediction errors of the angular velocity. The main innovation of this article is the addition of the yellow-line path, which provides an accuracy estimation of the moment of inertia via NN method. Figure 2 indicates that the network's input involves information from all the past observations, hence it can grasp long-term characteristics of the dynamical system. Therefore, the NN provides more precise estimation of the moment of inertia, so as to improve the estimation and prediction accuracy of attitude motion of the rotating space debris.

The rest of this article are organized as follows. Section 2 introduces the dynamical system of the rotating space debris. Based on the dynamical model, a NN and its associated preprocessing algorithm are proposed in Section 3. Then a nonlinear filter is designed in Section 4, which uses the NN to provide virtual observations of the moment of inertia. Furthermore, Section 5 provides some simulations to demonstrate performances of the novel filter in attitude estimation and prediction missions compared with the conventional EKF method. Finally, some conclusions are drawn in Section 6.

2 | DYNAMICAL MODEL

The aim of the attitude estimation problem (AEP) is to estimate the attitude quaternion $\boldsymbol{q}$ and angular velocity $\boldsymbol{\omega}$ of a rotating space debris using the noised measurements. In practice, it is sufficient to model the target as a rigid body, whose attitude dynamical equation is given by

$$\begin{aligned}\dot{q}_0 &= -\frac{1}{2}\boldsymbol{\omega}\cdot\boldsymbol{q}_v\\ \dot{\boldsymbol{q}}_v &= \frac{1}{2}\left[q_0\boldsymbol{\omega}+\boldsymbol{\omega}\times\boldsymbol{q}_v\right],\\ \dot{\boldsymbol{\omega}} &= \boldsymbol{J}^{-1}(\boldsymbol{J}\boldsymbol{\omega}\times\boldsymbol{\omega})+\boldsymbol{\varepsilon},\end{aligned} \tag{1}$$

where $\boldsymbol{q}=[q_0,\boldsymbol{q}_v]^T$ is the target's attitude quaternion with respect to the inertial coordinate frame, q_0 and $\boldsymbol{q}_v=[q_1,q_2,q_3]^T$ are the scale part and vector part of it, $\boldsymbol{\omega}=[\omega_x,\omega_y,\omega_z]^T$ is the target's angular velocity, $\dot{q}_0,\dot{\boldsymbol{q}}_v$, and $\dot{\boldsymbol{\omega}}$ are their corresponding time derivatives, $\boldsymbol{\varepsilon}$ is the disturbing torque acting on the target, and $\boldsymbol{J}$ is the target's inertia tensor in the form of

$$\boldsymbol{J}=\begin{bmatrix}J_{xx} & & \\ & J_{yy} & \\ & & J_{zz}\end{bmatrix}, \tag{2}$$

where J_{xx}, J_{yy}, and J_{zz} are the target's principal axes moments of inertia.

Obviously, the value $\boldsymbol{J}$ can influence the $\boldsymbol{\omega}$ and $\boldsymbol{q}$, hence one can identify the target's inertia tensor by measuring its angular velocity or attitude quaternion. In practice, it is usually hard to measure the noncooperative target's angular velocity or inertia characteristics directly, while the relative attitude could be acquired through noncontact

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measurement instruments, like Light Detection and Ranging (LIDAR) and vision systems . Therefore the observation equation is given by

$$\boldsymbol{z} = \boldsymbol{h}(\boldsymbol{q}, \boldsymbol{\omega}) + \boldsymbol{v} = \boldsymbol{q} + \boldsymbol{v}, \tag{3}$$

where $\boldsymbol{v}$ denotes the measurement noise.

Consequently, the inertia parameters identification of the tumbling target boils down to the problem about how to determine the values of $\boldsymbol{J}$ from the measurements of $\boldsymbol{q}$, based on the dynamical equations (1) and the observation equation (3).

3 | CHARACTERISTIC FREQUENCY INPUT NETWORK

3.1 | Discrete Fourier transformation: A frequency analysis method

As mentioned above, affects of the inertia parameters on the attitude motion is long term, thus a number of measurements are necessary to guarantee the accuracy of the inertia identification. However, a high dimension of input will result in the heavy computational burden and instability of the neural network in training process. Therefore, the measurement data must be compressed before sent to the neural network. In this section, the discrete Fourier transformation (DFT) method is utilized to exact the characteristic frequencies of the tumbling target's attitude quaternion, so as to reduce the input dimension.

The DFT technique is widely used in signal processing and information compression.[16,17] Specifically, the DFT of the measurement data, denoted by $\boldsymbol{Z}(n)$, is given by[18,19]

$$Z_i(n) = \sum_{k=0}^{N-1} z_i(k) \mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}nk} \quad (i = 0, 1, 2, 3), \tag{4}$$

where j is the symbol of imaginary, N is the amount of the measurement data, $z_i(k)$ is the ith component of $\boldsymbol{z}$ at the time index k, and $Z_i(n)$ is the corresponding DFT result in the frequency domain at the frequency $f = \frac{n-1}{N\Delta t}$. For a tumbling rigid body, there are three significant peaks in the frequency domain,[20,21] as shown in Figure 3 for instance. The corresponding frequencies of the three peaks are denoted by f_1, f_2, and f_3, respectively. Moreover, $p_{ij}(i = \{0, \dots, 3\}, j = \{1, 2, 3\})$ denotes the height of peak at the frequency f_j, for the component z_i in the frequency domain.

Then, a set of characteristic frequency parameters are exacted from the DFT results of the observation data, that is,

$$\boldsymbol{x}_c = [f_1, f_2, f_3, p_{01}, p_{02}, \dots, p_{33}]^T. \tag{5}$$

Namely, the information of measurement data are compressed into these 15 parameters.

The application of the characteristic frequency vector $\boldsymbol{x}_c$ results in four main advantages:

1) The information dimension is significantly reduced. The dimension of $\boldsymbol{x}_c$ is 15, while the number of the measurements is 200–500, with four dimensions for each measurement. 2) Most of the measurement information is retained in the 15 parameters, because the three predominate frequency peaks contain most of the information energy. One can easily rebuild an approximation of the frequency domain from $\boldsymbol{x}_c$, and then use inverse DFT algorithm to get an approximation of the original measurement data in time domain, with little information lost. 3) The 15 parameters are constant over time, hence the start time of the measurements will not influence their values. 4) It can filter out the measuring noise $\boldsymbol{v}$, because the information energy of the noises are evenly distributed in the frequency domain.

3.2 | Neural network design

The information compression of DFT make it possible to identify the inertia parameters $\boldsymbol{J}$ by using a BP neural network. Based on the characteristic frequencies and the BP neural network, in this section, a characteristic frequency input network (CFIN) is provided for inertia identification of a tumbling space target.

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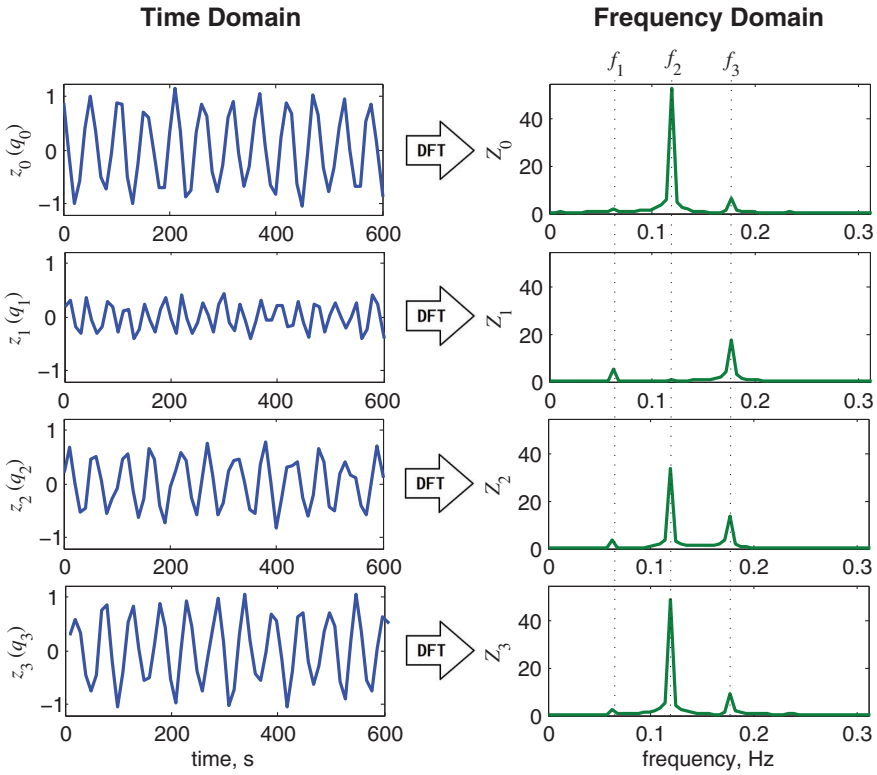


FIGURE 3 The discrete Fourier transformation (DFT) of attitude quaternion measurements

As shown in Figure 4, the red box represents the dynamical system of the target, where the inertia parameters $\boldsymbol{J}$ can influence the target's attitude quaternion and the angular velocity through the dynamical models (Equation 1). The yellow box represents the measuring process, where $\boldsymbol{z}(t)$ is the observation of $\boldsymbol{q}(t)$ with measurement noises. In the green box is the data preprocessing. The characteristic frequencies $\boldsymbol{x}_c$ are extracted from the measurements as mentioned above, and then be normalized for inputting to the neural network. And the blue box represents the BP neural network.

For the neural network, the inertia identification problem boils down to the parameter fitting problem. Then two layers are sufficient to obtain a good result,[22] including one hidden layer and one output layer. Moreover, the activation function of the hidden layer is the hyperbolic sine, while the output layer's is an identify function.

The number of nodes of the output layer should equal to the output dimension, that is, the degree of freedom of the inertia tensor $\boldsymbol{J}$. However, one can infer from Equation (1) that the norm of $\boldsymbol{J}$ does not influence the derivative $\dot{\boldsymbol{\omega}}$, because it is offset by the inverse $\boldsymbol{J}^{-1}$. Therefore one can only identify the inertia ratios between components of the inertia tensor, like J_{xx}/J_{yy}, through the quaternion measurements. Without loss of generality, we assume $J_{xx} < J_{yy} < J_{zz}$, and set the output as $\boldsymbol{y} = \left[\frac{J_{xx}}{J_{zz}}, \frac{J_{yy}}{J_{zz}}\right]^T$. These two ratios are the identifiable inertia parameters, whose values are in the interval $(0, 1]$. Therefore, two nodes are needed in the output layer.

The number of nodes of the hidden layer depends on the characteristics of the input data and the internal relation between input and output. In this article, it is selected as 100 through trial and error method. And the weights corresponding to these nodes are supposed to be trained using back propagation (BP) algorithm.

3.3 | Input preprocessing

In order to increase the learning speed of the neural network, the activation functions should work in the high-gain interval as much as possible.[22,23] It requires the input data to have a zero exception, and each component of input to has

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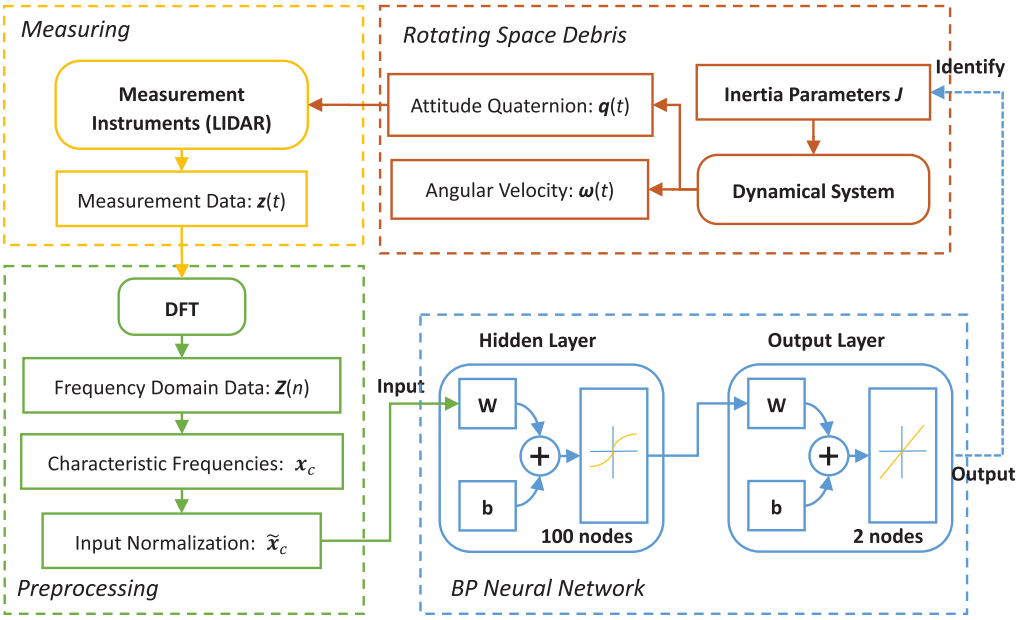


FIGURE 4 The process diagram of the neural network generating virtual observations of inertia parameters

a similar variance. Therefore, the normalization of the characteristic frequencies is necessary before they are entered to the BP neural network.

In this section, the input $\tilde{\boldsymbol{x}}_c$ is normalized using the mean value and the standard deviation of N_t training samples, that is,

$$\tilde{\boldsymbol{x}}_c = \boldsymbol{\sigma}^{-1}(\boldsymbol{x}_c - \boldsymbol{\mu}), \tag{6}$$

where

$$\boldsymbol{\mu} = \frac{1}{N_t}\sum_{i=0}^{N_t} \boldsymbol{x}_c^{(i)} \tag{7}$$

is the mean value of inputs of the training data, N_t is the number of training samples, $\boldsymbol{x}_c^{(i)}$ is the ith training sample, and $\boldsymbol{\sigma}$ is the Cholesky decomposition of the variance matrix of the training data, that is,

$$\boldsymbol{\sigma}\boldsymbol{\sigma}^T = \frac{1}{N_t - 1}\sum_{i=0}^{N_t} \left(\boldsymbol{x}_c^{(i)} - \boldsymbol{\mu}\right)\left(\boldsymbol{x}_c^{(i)} - \boldsymbol{\mu}\right)^T. \tag{8}$$

Moreover, the generation of the training samples are introduced in the next subsection.

3.4 | Network training

The neural network must be trained before it can work, which needs a batch of training samples. The number of training samples N_t depends on the number of nodes of the network and the complexity of the system for fitting. Specifically, the number of samples is set to be $N_t = 10{,}000$ for the CFIN through trial and error, which is an acceptable training scale for practical application.

It is impracticable to generate the training samples through physical experiments. Hence we get these data through simulation experiments based on the dynamical equations (1) and the observation equation (3). It is supposed that the

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target rigid body's mass distribution is uniformly random in the three-dimensional space, and the target's angular rate is 0.3–0.6 rad/s. Hence the process and parameters for generating one sample are provided as follows:

1) Generate a random vector $\boldsymbol{m} = [m_x, m_y, m_z]^T$, where m_x, m_y, and m_z are independent identically distributed (iid.) from the uniform distribution in the interval $[-1, 1]$. Then the principal moments of inertia of this sample are calculated by

$$J_{xx} = m_y^2 + m_z^2,\ J_{yy} = m_x^2 + m_z^2,\ J_{zz} = m_x^2 + m_y^2. \tag{9}$$

For the sake of convenience in the next analysis, exchange the values of them subject to $J_{xx} < J_{yy} < J_{zz}$.

2) Generate a random vector $\boldsymbol{\omega}_r = [\omega_{rx}, \omega_{ry}, \omega_{rz}]^T$, where ω_{rx}, ω_{ry}, and ω_{rz} are iid. from the uniform distribution in the interval $[-1, 1]$. Generate a random ω_{0r} which obeys the uniform distribution between 0.3 and 0.6. Then the initial angular velocity is set as

$$\boldsymbol{\omega}_0 = \omega_{0r} \frac{\boldsymbol{\omega}_r}{\|\boldsymbol{\omega}_r\|}. \tag{10}$$

3) Since the initial quaternion $\boldsymbol{q}_0$ does not influence the frequency parameters $\boldsymbol{x}_c$, it is set as $\boldsymbol{q}_0 = [1, 0, 0, 0]^T$ for the sake of convenience. Then one can generate the time series $\boldsymbol{q}(t)$ by utilizing the dynamical model equation (1).

4) Generate 500 measurements $\{\boldsymbol{z}(1), \ldots, \boldsymbol{z}(500)\}$ using the observation model equation (3), with measuring rate of 1 Hz, $\boldsymbol{v}$ modeled as a zero-mean Gaussian noise with variance of 0.05^2.

5) Using DFT to the 500 measurements to get the characteristic frequency vector $\boldsymbol{x}_c^{(i)}$ and normalize it using the preprocessing technique above.

6) Set the normalized vector $\tilde{\boldsymbol{x}}_c^{(i)}$ as the ith input, and the inertia ratios $\boldsymbol{y}^{(i)} = \left[\frac{J_{xx}}{J_{zz}}, \frac{J_{yy}}{J_{zz}}\right]^T$ as the ith output.

By repeating the six steps above, one can generate 10,000 iid. samples. Then, the samples are divided into three parts: 8000 samples for training, 1500 for validation, and 500 for testing. The training samples are presented to the neural work during training, and the network is adjusted according to its error. The validation samples are used to measure network generalization, and to halt training when generalization stops improving. The testing samples have no effect on training and so provide an independent measure of network performance during training. Moreover, another 2000 samples are generated independently to test the CFIN performance after training.

The aim of training is to reduce the mean square error (mse) of outputs of the neural network, which is defined as

$$\text{mse} = \sum_{k=1}^{N_s} \sum_{m=1}^{2} (y_{k,m,out} - y_{k,m,true})^2, \tag{11}$$

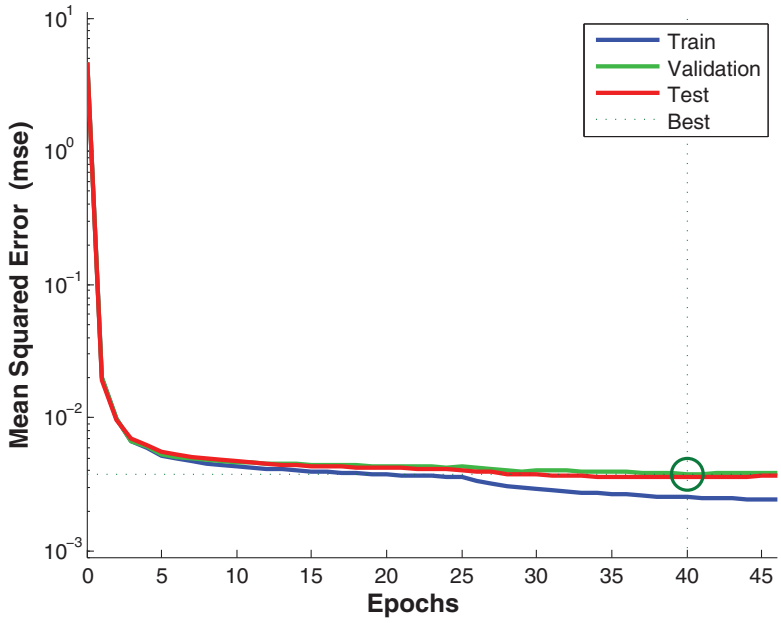


FIGURE 5 Performances of the neural network in training

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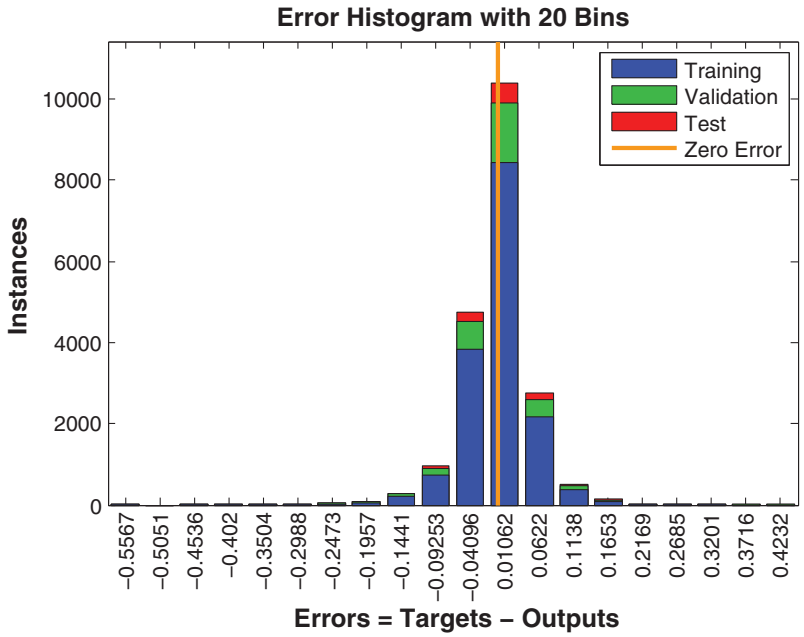


FIGURE 6 Distribution of output errors of the trained network

where N_s denotes the number of samples, $y_{k,m,out}$ denotes the mth element of output for the kth sample input, and $y_{k,m,true}$ denotes the corresponding label of the kth sample (i.e., the exception of the output $y_{k,m,out}$).

The Levenberg–Marquardt algorithm[23] is applied for training the neural network. And the train automatically stops when generation no longer improves after 40 epochs, which is indicated by an increase in the mean square error of the validation samples, as shown in Figure 5. Moreover, Figure 6 shows the distribution of output errors of the neural network. It is illustrated that in most cases the output errors are less than 0.05, which is sufficiently accurate for attitude estimation of the rotating space debris.

4 | NONLINEAR FILTER FOR PARAMETER ESTIMATION

4.1 | Observation equations

By extracting dynamical parameters from Equation (1), the state vector of the rotating space debris is written as

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{q}_v \\ \boldsymbol{\omega} \\ \boldsymbol{y} \end{bmatrix}, \tag{12}$$

where $\boldsymbol{y} = \left[\frac{J_{xx}}{J_{zz}}, \frac{J_{yy}}{J_{zz}}\right]^T$ is the inertia ratio as defined in Section 3.

Note that the attitude quaternion has only three degrees of freedom. For the purpose to reduce the dimension of the system, the vector $\boldsymbol{x}$ does not contain the quaternion's scalar part q_0 which is redundancy for the state.

Assume that the LIDAR or vision system gives a noisy measurement of the attitude quaternion $\boldsymbol{q}$ of the space target with respect to the inertial coordinate frame. Moreover, the neural network proposed in Section 3 provides a virtual observation of the target's inertia ratios $\boldsymbol{y}$. Then the measurement vector in Equation (3) is updated to

$$\boldsymbol{z} = \boldsymbol{h}(\boldsymbol{x}) + \boldsymbol{v} = \begin{bmatrix} \boldsymbol{q} \\ \boldsymbol{y} \end{bmatrix} + \boldsymbol{v}, \tag{13}$$

where $\boldsymbol{v}$ is the measurement noise, which is assumed to be white with covariance $\boldsymbol{R} = E[\boldsymbol{v}\boldsymbol{v}^T]$.

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Then the observation sensitivity matrix is obtained by calculate the derivative of the observation vector $\boldsymbol{z}$ with respect to the state $\boldsymbol{x}$ as

$$\boldsymbol{H} = \begin{bmatrix} \boldsymbol{I}_{4\times4} & \boldsymbol{0}_{4\times3} & \boldsymbol{0}_{4\times2} \\ \boldsymbol{0}_{2\times4} & \boldsymbol{0}_{2\times3} & \boldsymbol{I}_{2\times2} \end{bmatrix}, \tag{14}$$

where $\boldsymbol{I}_{n\times n}$ denotes the n-by-$n$dimensional identity matrix, and $\boldsymbol{0}_{n\times n}$ denotes the n-by-n dimensional zero matrix.

4.2 | Extended Kalman filter design

As mentioned above, absolute values of inertia parameters do not effect the dynamics of the rotating space target. Without loss of generation, one can assume the biggest moment of inertia as $J_{zz} = 1$ in the filter process. Then considering the definition of $\boldsymbol{y} = \left[\frac{J_{xx}}{J_{zz}}, \frac{J_{yy}}{J_{zz}}\right]^T$, the target's inertia tensor is given by

$$\boldsymbol{J} = \begin{bmatrix} y_1 & & \\ & y_2 & \\ & & 1 \end{bmatrix}, \tag{15}$$

where y_n denotes the nth component of state vector $\boldsymbol{y}$.

Then by substituting Equation (15) into Equation (1), one can obtain the compact form of the process dynamics as

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} \frac{1}{2}(\omega_x q_0 + \omega_z q_2 - \omega_y q_3) \\ \frac{1}{2}(\omega_y q_0 - \omega_z q_1 + \omega_x q_3) \\ \frac{1}{2}(\omega_z q_0 + \omega_y q_1 - \omega_x q_2) \\ \frac{y_2-1}{y_1}\omega_y\omega_z \\ \frac{1-y_1}{y_2}\omega_x\omega_z \\ (y_1 - y_2)\omega_x\omega_y \\ \boldsymbol{0}_{2\times1} \end{bmatrix} + \begin{bmatrix} \boldsymbol{0}_{3\times1} \\ \boldsymbol{\varepsilon} \\ \boldsymbol{0}_{2\times1} \end{bmatrix}. \tag{16}$$

In this article, the state transition matrix of the linearized system is needed for covariance propagation of the EKF. Considering a small variation of the attitude quaternion $\delta\boldsymbol{q} = \left[q_0,\ \boldsymbol{q}_v^T\right]^T$, one can linearize the dynamical equations of Equation (16) using the linearization technique[24,25] as

$$\delta\dot{q}_0 = 0, \tag{17}$$

$$\delta\dot{\boldsymbol{q}}_v = -\boldsymbol{\omega} \times \delta\boldsymbol{q}_v + \frac{1}{2}\delta\boldsymbol{\omega}. \tag{18}$$

Moreover, linearization of $\dot{\boldsymbol{\omega}}$ in Equation (16) yields

$$\left.\frac{\partial\delta\dot{\boldsymbol{\omega}}}{\partial\delta\boldsymbol{\omega}}\right|_{\boldsymbol{x}=\hat{\boldsymbol{x}}} = \left.\frac{\partial\dot{\boldsymbol{\omega}}}{\partial\boldsymbol{\omega}}\right|_{\boldsymbol{x}=\hat{\boldsymbol{x}}} = \boldsymbol{M}(\hat{\boldsymbol{\omega}}, \hat{\boldsymbol{y}}) = \begin{bmatrix} 0 & \frac{\hat{y}_2-1}{\hat{y}_1}\hat{\omega}_z & \frac{\hat{y}_2-1}{\hat{y}_1}\hat{\omega}_y \\ \frac{1-\hat{y}_1}{\hat{y}_2}\hat{\omega}_z & 0 & \frac{1-\hat{y}_1}{\hat{y}_2}\hat{\omega}_x \\ (\hat{y}_1 - \hat{y}_2)\hat{\omega}_y & (\hat{y}_1 - \hat{y}_2)\hat{\omega}_x & 0 \end{bmatrix}, \tag{19}$$

$$\left.\frac{\partial\delta\dot{\boldsymbol{\omega}}}{\partial\delta\boldsymbol{y}}\right|_{\boldsymbol{x}=\hat{\boldsymbol{x}}} = \left.\frac{\partial\dot{\boldsymbol{\omega}}}{\partial\boldsymbol{y}}\right|_{\boldsymbol{x}=\hat{\boldsymbol{x}}} = \boldsymbol{N}(\hat{\boldsymbol{\omega}}, \hat{\boldsymbol{y}}) = \begin{bmatrix} -\frac{\hat{y}_2-1}{\hat{y}_1^2}\hat{\omega}_y\hat{\omega}_z & \frac{1}{\hat{y}_1}\hat{\omega}_y\hat{\omega}_z \\ -\frac{1}{\hat{y}_2}\hat{\omega}_x\hat{\omega}_z & -\frac{1-\hat{y}_1}{\hat{y}_2^2}\hat{\omega}_x\hat{\omega}_z \\ \hat{\omega}_x\hat{\omega}_y & -\hat{\omega}_x\hat{\omega}_y \end{bmatrix}, \tag{20}$$

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where $\hat{\boldsymbol{x}}$ denotes the estimation value of the state vector $\boldsymbol{x}$.

Then the process sensitivity matrix of the dynamical system is given by

$$\boldsymbol{F} = \left.\frac{\partial \delta \dot{\boldsymbol{x}}}{\partial \delta \boldsymbol{x}}\right|_{\boldsymbol{x}=\hat{\boldsymbol{x}}} = \begin{bmatrix} -[\boldsymbol{\omega}\times] & \frac{1}{2}\boldsymbol{I}_{3\times 3} & \boldsymbol{0}_{3\times 2} \\ \boldsymbol{0}_{3\times 3} & \boldsymbol{M}(\hat{\boldsymbol{\omega}}, \hat{\boldsymbol{y}}) & \boldsymbol{N}(\hat{\boldsymbol{\omega}}, \hat{\boldsymbol{y}}) \end{bmatrix}. \tag{21}$$

Furthermore, the equivalent discrete-time system of the linearized system is written as

$$\delta \boldsymbol{x}_{k+1} = \boldsymbol{\Phi}_k \delta \boldsymbol{x}_k + \boldsymbol{w}_k, \tag{22}$$

where $\boldsymbol{w}_k$ is the discrete-time process noise, and the first-order state transition matrix $\boldsymbol{\Phi}_k$ at time t_k over the time interval $\Delta t = t_{k+1} - t_k$ is set as

$$\boldsymbol{\Phi}_k = \boldsymbol{I}_{8\times 8} + \Delta t \boldsymbol{F}(t_k). \tag{23}$$

Moreover, the covariance of the discrete-time process noise $\boldsymbol{w}$ is related to the continuous process noise covariance as

$$\boldsymbol{Q}_k = E\{\boldsymbol{w}_k \boldsymbol{w}_k^T\} = \mathrm{diag}\left(\boldsymbol{0}_{3\times 3}, \Delta t E\{\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^T\}, \boldsymbol{0}_{2\times 2}\right), \tag{24}$$

where $E\{\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^T\}$ denotes the covariance of disturbing torques acting on the space debris.

Now, an EKF estimator can be designed based on the derived models (14), (23), and (24). The EKF algorithm for estimating the space debris' attitude motion and inertia parameters is given in two steps:

1) Estimate propagation

$$\hat{\boldsymbol{x}}_{k/k-1} = \hat{\boldsymbol{x}}_{k-1} + \int_{t_{k-1}}^{t_k} \boldsymbol{f}(\boldsymbol{x}) \mathrm{d}t, \tag{25}$$

$$\boldsymbol{P}_{k/k-1} = \boldsymbol{\Phi}_k \boldsymbol{P}_{k-1} \boldsymbol{\Phi}_k^T + \boldsymbol{Q}_k. \tag{26}$$

2) Estimate update

$$\boldsymbol{K}_k = \boldsymbol{P}_{k/k-1} \boldsymbol{H}_k^T (\boldsymbol{H}_k \boldsymbol{P}_{k/k-1} \boldsymbol{H}_k^T + \boldsymbol{R}_k)^{-1}, \tag{27}$$

$$\boldsymbol{P}_k = (\boldsymbol{I}_{8\times 8} - \boldsymbol{K}_k \boldsymbol{H}_k) \boldsymbol{P}_{k/k-1}, \tag{28}$$

$$\Delta \boldsymbol{x}_k = \boldsymbol{K}_k (\boldsymbol{z}_k - \boldsymbol{h}(\hat{\boldsymbol{x}}_{k/k-1})), \tag{29}$$

where $\hat{\boldsymbol{x}}_{k/k-1}$ and $\hat{\boldsymbol{x}}_k$ are priori and posteriori estimates of the state vector, and $\boldsymbol{P}_k$ is the estimate of error covariance of $\hat{\boldsymbol{x}}_k$. The update of $\hat{\boldsymbol{\omega}}_k$ and $\hat{\boldsymbol{y}}_k$ can be easily obtained by adding the corresponding error to the priori estimation. However, the update of $\hat{\boldsymbol{q}}$ is more complicated because only the vector part of the quaternion error $\Delta \hat{\boldsymbol{q}}_{v,k}$ is given at every step time. Therefore, first, the scalar part of the quaternion error should be computed from the corresponding vector parts as

$$\Delta \hat{q}_{0,k} = \sqrt{1 - \hat{\boldsymbol{q}}_v^T \hat{\boldsymbol{q}}_v}. \tag{30}$$

Second, the quaternion are updated using the quaternion multiplication rule. Consequently, the state update process after Equation (29) is given by

$$\hat{\boldsymbol{q}}_k = \hat{\boldsymbol{q}}_{k/k-1} \circ \begin{bmatrix} \Delta \hat{q}_{0,k} \\ \Delta \hat{\boldsymbol{q}}_{v,k} \end{bmatrix}, \tag{31}$$

$$\hat{\boldsymbol{\omega}}_k = \hat{\boldsymbol{\omega}}_{k/k-1} + \Delta \hat{\boldsymbol{\omega}}_k, \tag{32}$$

$$\hat{\boldsymbol{y}}_k = \hat{\boldsymbol{y}}_{k/k-1} + \Delta \hat{\boldsymbol{y}}_k, \tag{33}$$

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where the circle "∘" denotes the quaternion multiply operator defined by

$$\begin{bmatrix} a_0 \\ \boldsymbol{a}_v \end{bmatrix} \circ \begin{bmatrix} b_0 \\ \boldsymbol{b}_v \end{bmatrix} = \begin{bmatrix} a_0 b_0 - \boldsymbol{a}_v \boldsymbol{b}_v \\ a_0 \boldsymbol{b}_v + b_0 \boldsymbol{a}_v - \boldsymbol{a}_v \times \boldsymbol{b}_v \end{bmatrix}. \tag{34}$$

5 | SIMULATION STUDIES

In this section, extensive Monte Carlo simulations are introduced to illustrate the performance of the proposed NNEKF. In the simulations, the space debris is assumed an inactive uncontrolled satellite tumbling in space (i.e., the orientation of the angular velocity does not coincide with any inertial principal axis). And the dynamical parameters of the target are absolutely unknown. The moments of inertia of the target are chosen to be $I_{xx} = 400$ kg·m^2, $I_{yy} = 500$ kg·m^2, and $I_{zz} = 600$ kg·m^2, which are not a priori known in simulations. Therefore the truth values of inertia ratios are $y_1 = \frac{I_{xx}}{I_{zz}} = 0.667$ and $y_2 = \frac{I_{yy}}{I_{zz}} = 0.833$.

In simulations, the initial attitude quaternion is set as $\boldsymbol{q}(0) = [1, 0, 0, 0]^T$, and the initial angular velocity is set as $\boldsymbol{\omega}(0) = [0.11, 0.13, 0.18]^T$ rad/s. In addition, the effects of disturbance torques acting on the target are modeled as zero-mean Gaussian noises added on the angular acceleration with covariance $E\{\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T\} = 0.1^2\boldsymbol{I}_{3\times3}$ (N·m)^2.[26]

It is initially assumed that the quaternion-expression attitude measurements of the space debris are provided by noncontact measure methods, denoted by $\{\boldsymbol{z}_k\}$. The measurement errors are also assumed zero-mean Gaussian noises with covariance $\boldsymbol{R}_k = E\{\boldsymbol{z}_k\boldsymbol{z}_k^T\} = 0.05^2\boldsymbol{I}_{4\times4}$. In addition, neither the angular velocity nor the moment of inertia is measurable.

In order to demonstrate performances of the proposed NNEKF, 50 trials of Monte Carlo simulations are implemented using the NNEKF and the conventional EKF, respectively. For the sake of convenience, all parameters used in simulations are listed in Table 1.

In each Mont Carlo simulation, the NNEKF and the conventional EKF are implemented to estimate the rotating space debris' dynamical parameters (i.e., the attitude quaternion, the angular velocity, and the inertia ratios) in the time interval 0–600 s. The NNEKF uses the neural network to get an estimation of the inertia ratios, based on the original measurements at the time interval between 0 and 300 s (i.e., the amount of measurements is set as $N = 300$ in Equation 4). Namely, the NNEKF does not work in the first 300 s until the inertia ratios are well estimated. At the time $t = 300$ s, the NNEKF firstly filters the past measurements, and then deals with the current measurement in real time. The fourth-order Runge–Kutta method is utilized to predict the space debris' attitude motion when the measurement no longer updates in the time interval 600–800 s, based on the last estimates of the dynamical parameters at the time $t = 600$ s.

Figure 7 shows the estimation and prediction errors of the attitude quaternion for a typical run in simulations. The figure shows that the standard deviation of measurement errors is about 0.1 for each component of the attitude

TABLE 1 Parameters used in simulations

Parameter	Denotation	Value
Inertial tensor of the chaser satellite	$\boldsymbol{J}$	diag(400,500, 600) kg·m^2
Covariance of disturbing moments act on the target	$E\{\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T\}$	$0.1^2\boldsymbol{I}_{3\times3}$ (N · m)^2
Measurement error covariance	$\boldsymbol{R}$	$0.05^2\boldsymbol{I}_{4\times4}$
Measurement frequency	$1/\Delta t$	1 Hz
Estimation time	t_e	0–600 s
Prediction time	t_p	600–800 s
Initial quaternion of the target	$\boldsymbol{q}(0)$	$[1,\ 0,\ 0,\ 0]^T$
Initial angular velocity	$\boldsymbol{\omega}_t(0)$	$[0.11,\ 0.13,\ 0.18]^T$ rad/s
Initial estimate of the state vector	$\hat{\boldsymbol{x}}_0$	$\boldsymbol{0}_{1\times8}$
Initial error covariance matrix of estimates	$\boldsymbol{P}_0$	$\boldsymbol{I}_{8\times8}$

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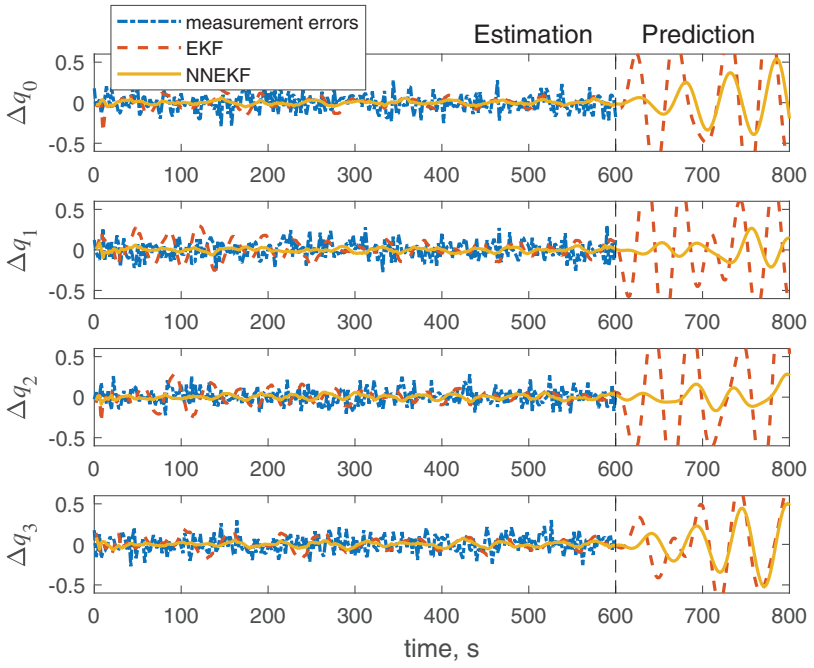


FIGURE 7 Attitude quaternion errors of the NNEKF and conventional EKF

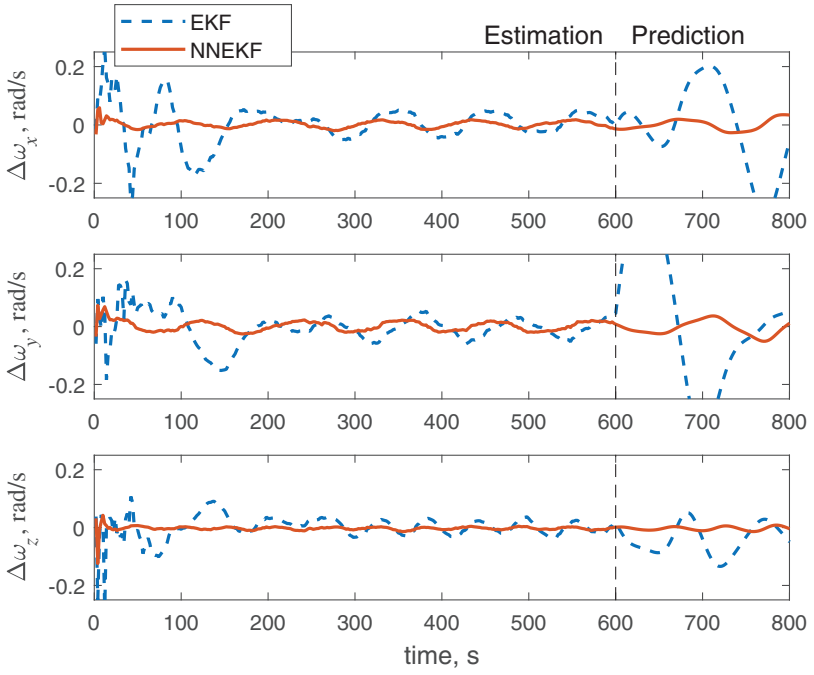


FIGURE 8 Angular velocity errors of the NNEKF and conventional EKF

quaternion. By considering the propagation of dynamical parameters, both the NNEKF and the conventional EKF reduce the quaternion errors to a certain extent. However, since the estimates of inertia ratios are not accuracy, the prediction errors accumulate very fast when no measurement is available to update the state. Figure 7 shows that the prediction errors of the quaternion rapidly diverge to more than 0.5 in about 50 s when using estimates of the conventional EKF method. Meanwhile, the prediction errors accumulate relatively slowly when using the NNEKF method, which can provide more accuracy estimates of the inertia ratios.

Figure 8 shows the estimation and corresponding prediction errors of the angular velocity when using the proposed NNEKF and the conventional EKF method. It is indicated that owing to virtual observations of inertia ratios, the NNEKF

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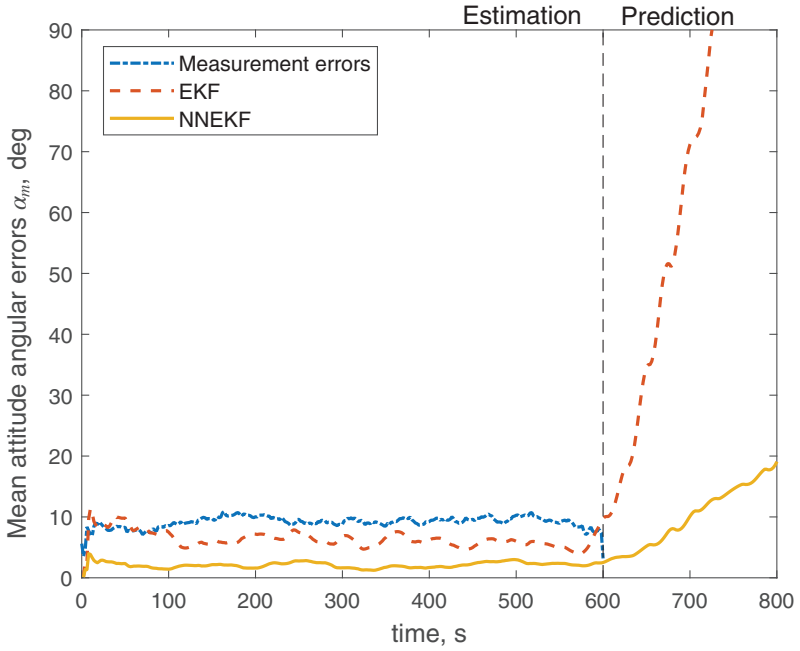


FIGURE 9 Mean attitude angular errors of the NNEKF and conventional EKF in 50 Monte Carlo runs

provides more accuracy estimates of the angular velocity than the conventional EKF method. Moreover, the angular velocity's prediction errors suggest to be very stable when using the estimates of the NNEKF, which diverge very rapidly when using the conventional EKF .

To indicate the estimate accuracy more distinctly, the attitude error angle α is introduced to represent the angle between the estimated body-fixed frame (calculated from the estimates of the quaternion) and the truth body-fixed frame of the space debris, defined as:

$$\alpha = 2\arccos\sqrt{\Delta\boldsymbol{q}_v^T\Delta\boldsymbol{q}_v}, \tag{35}$$

where $\Delta\boldsymbol{q} = \hat{\boldsymbol{q}}^{-1}\circ\boldsymbol{q}$ is the error quaternion between the estimate $\hat{\boldsymbol{q}}$ and the truth value $\boldsymbol{q}$, and $\Delta\boldsymbol{q}_v$ denotes the vector part of it.

Then the Monte Carlo mean attitude angular error is calculated as

$$\alpha_{m,k} = \frac{1}{50}\sum_{n=1}^{50}\alpha_{(n,k)}, \tag{36}$$

where $\alpha_{(n,k)}$ denotes the attitude error corresponding to estimation/prediction of the state vector at the time index $k$in the nth run of Monte Carlo simulations.

Figure 9 shows the mean attitude angular errors of the NNEKF and the conventional EKF in 50 Monte Carlo runs. It is shown that the mean measurement error is about 10 degrees. The estimation error using the conventional EKF is about 6.1 degrees at the time interval 400–600 s. And the estimation error using the NNEKF is 2.4 degrees, which is only 39% as the conventional EKF. Moreover, the prediction errors of the conventional EKF rapidly diverge to more than 90 degrees in about 120 s, while the prediction errors of the NNEKF just increase to about 20 degrees in 200 s. Generally, the accumulate rate of prediction errors of the NNEKF is just about 24% as the conventional EKF.

6 | CONCLUSIONS

In this article, a neural-network-enhanced Kalman filter (NNEKF) is proposed to improve the precision and robustness of attitude estimation and prediction of rotating space debris. The main innovation of the NNEKF is to utilize virtual

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observations of inertia characteristics to improve the filter's performance. In order to generate virtual observations of inertia ratios, a neural network is designed and offline trained using simulation data. Moreover, the input data of the network are preprocessed using the discrete Fourier transformation (DFT) method, so as to decrease the number of nodes of the network and furthermore reduce the computational burden in training it. By involving the characteristic frequencies in the input vector, the neural network can extract information from all the past observations, so as to grasp long-term characteristics of the dynamical system. Therefore, the NNEKF can provide more precise estimation of the target's moment of inertia, and furthermore improve the accuracy and robustness of attitude estimation and prediction. A series of Monte Carlo simulations are implemented to demonstrate performances of the proposed novel method. Simulation results indicate that the NNEKF can reduce the estimation errors by 39% compared with the conventional EKF method when using the same measurement data. And the accumulation errors of prediction using estimates of the NNEKF is just as 24% as the conventional EKF.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Ma C, Zheng Z, Chen J, Yuan J. Robust attitude estimation of rotating space debris based on virtual observations of neural network. *Int J Adapt Control Signal Process*. 2022;36(2):300-314. https://doi.org/10.1002/acs.3297

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